

Organizational Ethos, Responsibilities, and Challenges**Overview**

10 Feb 2026

- doing ethics on the ground is challenging ('specially when you are new)
- how can data scientists create a responsible workflow?
- practice what operationalizing ethics looks like

Readings

It's about power: What ethical concerns do software engineers have, and what do they (feel they can) do about them?

<https://doi.org/10.1145/3593013.3594012>

Walking the Walk of AI Ethics: Organizational Challenges and the Individualization of Risk among Ethics Entrepreneurs

<https://doi.org/10.1145/3593013.3593990>

Data Science Ethos Lifecycle: Interplay of Ethical Thinking and Data Science Practice

<https://doi.org/10.1080/26939169.2022.2089411>

Values Levers: Building Ethics into Design

<https://doi.org/10.1177/0162243912436985>

Near-Term Artificial Intelligence and the Ethical Matrix

<https://doi.org/10.1093/oso/9780190905033.003.0009>

Summaries**It's About Power**

- this paper **investigates** how software **engineers identify** and act on **ethical concerns** in their work
- the authors surveyed **115 engineers** and interviewed **21 practitioners** across **organizational contexts** to capture self-identified **ethical concerns**
- **reported concerns**
 - **privacy**
 - **surveillance**
 - **military** uses
 - **questions** about the **purpose** and **impact** of their **work**
- **engineers** in relatively **powerful** positions often **felt unable** to **resolve** concerns due to **organizational incentives**, personal **precarity**, and a **lack** of collective **power**

	study concludes that ethical interventions should move beyond helping engineers identify issues to supporting their capacity to collectively influence decisions and organizational priorities
Walking the Walk of AI Ethics	- this article examines how AI ethics values are actually implemented within tech firms authors find that ethics work is structurally undermined “decoupling” – when formal policies do not translate into practice ethics champions (ethics entrepreneurs) – struggle to prioritize ethics in environments focused on product and business metrics ethical work is hard to quantify, and frequent organizational restructuring impedes continuity individuals bear significant personal risk when raising ethical concerns (especially those from marginalized groups) - study reveals that ethical AI work inside firms often depends on individual labor rather than systemic support
Data Science Ethos Lifecycle	- his article introduces the Data Science Ethos Lifecycle, a conceptual and pedagogical framework that embeds ethical reflection directly into the stages of data science work - the framework encourages practitioners and students to integrate social context and consequences at each stage of the data science lifecycle four conceptual lenses to highlight how choices in data science shape and are shaped by societal structures positionality power sociotechnical systems narratives Ethos Lifecycle connects ethical theory to practical workflows, aiming to equip practitioners with tools to articulate and act on their responsibilities to diverse stakeholders
Values Levers	Katie Shilton explores how ethics can be integrated into design practices rather than treated as an external add-on - the author identifies design activities that function as values levers through an ethnographic study of a laboratory developing mobile software

	- these levers include structured activities, conversations among designers, and interventions by mentors or social scientists that create space for ethical reflection and consensus around design values - by activating such levers, design practice itself can foreground social values such as privacy and anti-surveillance, encouraging ethical action to become part of routine engineering work
Near-Term Artificial Intelligence and the Ethical Matrix	- addresses ethical reflection about near-term AI developers and deployers bear ethical responsibility for their societal impacts of AI systems already deployed in sectors like advertising, credit scoring, and criminal justice - propose ethical matrix as a tool for systematic reflection during design and deployment to consider how various stakeholders' interests are affected across key ethical dimensions wellbeing autonomy fairness transparency - reframes AI ethics from abstract philosophical discourse to a practical tool used by practitioners during the development process, emphasizing accountability and stakeholder inclusion
Key Takeaways	
Ethical concerns are broad and grounded in practice	in It's about power, software engineers report concerns ranging from privacy and surveillance to the very purpose of their work. These concerns are not abstract but tied to concrete technological decisions and social impacts. Importantly, engineers feel constrained by organizational incentives, suggesting that ethics cannot be decoupled from institutional context
Organizational structures shape ethical outcomes	walking the Walk of AI Ethics shows that formal ethics policies rarely translate into sustained practice. Staff designated with ethical responsibilities often lack authority, metrics, or structural support, leading to ethical work being uneven, risky, and highly individual. This highlights a disconnect between ethics talk and ethics practice

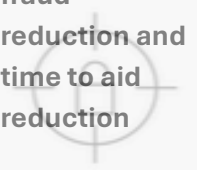
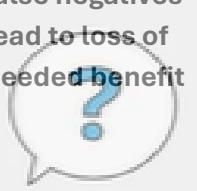
Embedding ethics requires methodological frameworks	the Data Science Ethos Lifecycle provides a practical framework integrating ethical reflection at each stage of data work, emphasizing lenses such as power and positionality. This model bridges scholarly ethical thinking and the everyday workflow of practitioners
Design practices can be levers for ethical reflection	Values Levers reveals that specific design practices – structured discussions, role negotiations, and value articulation – can bring ethics into early design phases, shaping how technologies embody social values
Tools for ethical assessment support accountability	the Ethical Matrix framework offers a structured approach for reflecting on impacts of near-term AI across stakeholders and values (e.g., autonomy, fairness, wellbeing), making ethical analysis actionable during development
Structural Ethical Practice	ethics must be integrated into organizational incentives and workflows so engineers and designers can act meaningfully. - frameworks and tools (like the Ethos Lifecycle and ethical matrix) are necessary to operationalize reflection and accountability. - organizational and social context matters—ethical decision-making cannot be isolated from institutional power structures and metrics.
Async	
Top Concerns and Challenges	
Interview Key Stakeholders	the best way to ‘figure out’ the ethical concerns and dominant narratives around data science is direct communication
University Study (“Colleges”)	- employed open-ended surveys found that there were common concerns - how their work might be used in a military setting privacy violations - was collected data being sold to advertisers - surveillance, corporate surveillance, government surveillance environmental implications - impact on labor market - not enough time put into code maintenance

Ethical Entrepreneurs	- increasingly becoming a role in data science - from all fields in various roles - often part of teams dedicated to issues like trust, safety, civic engagement, privacy and security - often folded into larger companies who can afford ethical sounding boards - those tasked with upholding ethical standards often encountered roadblocks - frequent shakeups often related to Agile development methods can present challenges when trying to maintain consistent ethical accountability ethical concerns can often be deprioritized or ignored relative to “larger” product goals - there can be a psychological toll that can come from constantly raising awareness of ethical issues if you are in an organization that does not necessarily value those insights
Ethos Lifestyle	
Privilege Hazard	- those who occupy the most privilege in society are often tasked with shaping ethical standards - those who hold positions of power are often poorly equipped to recognize instances of oppression and are rarely held accountable for ethical violations
Integrating Ethical Reasoning Throughout the Product Lifecycle	- data science emerges from and shapes the world - there are many lenses through which data scientists can consider the consequences of their work positionality – you can inform a person’s relative social circumstances by thinking of how concepts like race, gender, class, religion, culture, education, or ableism intersect the role of power – often symmetrical and uneven allowing inequality to flow opaquely throughout the sociotechnical systems that we create sociotechnical systems describe how technology and people are intertwined, and considers how the tools we create might be co-produced, co-opted, or even replace existing systems

Questions that a data scientists might ask in relation to who they work for	• what kind of data is going to be collected? • who will have access to the data? • what are the conditions by which the data will be stored and sold? • if it is collected in a way that a person does not know or understand, what might that informed consent process look like?
Sociotechnical Prompts	• thinking about the interrelation between technology and society by questioning assumptions that are embedded into the data ○ how do my tools enable or constrain action? ○ who will be held responsible if something goes wrong? ○ does my project contribute to or modify an existing sociotechnical system? ○ is it seeking to replace it entirely? ○ what other systems are needed or need to be built after my tool is introduced into the market? ○ how might this system be exploited in ways that are unintentional or unanticipated?
Value Levers	
Designers Values	• the values held by the designers inevitably impact how ○ the products that they develop ○ information technology and data science is imagined ○ data is handled and stored ○ designers create the categories that tag the data • there are pressures within the tech industry to innovate quickly • values-oriented design processes can be seen as unattractive in an environment that prized scalability and efficiency
Areas Addressed by Value Levers	• internal prototyping of a product ○ how will the data be collected and stored? ○ are there ethical concerns? ○ one way to discover if the system might violate ethical values of the designers ○ allows system creators and designers to focus on how the data will be collected and stored ○ allows creators to focus on possible inferences that could be drawn from the data

	• creation of interdisciplinary teams ○ when members of a design team come from variety of fields, they bring to the table different narratives and they help to better understand how data will be represented, displayed, accessed, and used • internal support • funding
Power Structures	
Ethical Matrix	• allows people to engage with the ethical implication of technology without necessarily having to solve the problem • takes power into play by allowing people to engage with the ethical implications of technology • designed to let you be the stakeholder and consider how technology might impact your life
Who are the Stakeholders?	• based on who will be impacted • anyone deserving of any kind of moral or ethical consideration
Choosing Stakeholders	• decide who deserves consideration • capture each stakeholder's unique set of interests or concerns • apply ethical matrix to a setting

Ethical Matrix		Government	Families that Qualify for Assistance
Well-being	• How will each stakeholder benefit? • How might they each be harmed? • Are there less risky alternatives that could provide the same outcome?	by identifying people who do not need assistance, money is saved	harmed if benefits are denied or benefits are delayed due to staff shortages
Autonomy	• Will each stakeholder have a choice to use or be subject of the use? • Will they be able to determine how it is used with respect to their interests? • What will informed consent look like?	how adapting new technologies affects the workforce	does the digital divide create an obstacle

Justice	• Does the algorithm unfairly favor the interest of one stakeholder over another? • Do false negatives/positives harm or benefit one stakeholder over another?	fraud reduction and time to aid reduction 	false negatives lead to loss of needed benefit 
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ADSA Data Science Ethos	
The Lenses	• Conceptual lenses offer structured ways of thinking about the social and ethical contexts relevant to each stage of the data science research process • Each lens represents a theoretical stance connected to specific ethical problems • Four Lenses ○ Positionality ○ Sociotechnical Systems ○ Power ○ Narratives
The Stages	• question • discovery • analysis • modeling • interpreting • sharing
Case Studies	• estimating population with phone records • unfair tax practices in major U.S. cities • Crowd The Tap: protecting tap water
Estimating Population with Phone Records	
Summary	• enumerations of populations and their economic status are collected for many essential purposes ○ determining electoral representation ○ allocating tax dollars ○ deploying public health interventions • census are performed to collect this data, but they are expensive, time consuming, and infrequent •
Positionality	• use lens of positionality to examine the Data Discovery stage of a project using mobile phone records to estimate population size and economic status • when researchers work with sensitive data (ex: phone records) the data is obfuscated to protect individual identities

	• what culturally derived assumptions are being made? ○ for a culture that prized individual liberties over collective well-being, the priority data protection effort would logically be assumed to be the protection of identities while overlooking the harm that can be done to segments of the population if their locations are revealed <i>WHAT?</i>
Power	• in the case of using mobile phone records to estimate population size, location, and economic status, it is essential to remember during the Data Discovery stage that for-profit businesses have the power to decide what data they will share, with whom, and under what condition • if a company makes their phone records freely available to researchers, what is their motivation? altruism? pride? financial or political gain? • can the data be trusted?
Sociotechnical Systems	• how are the systems that generated the data intertwined with the lives of real people • in some regions a phone number uniquely identifies a single person while other regions a single phone can be shared by a whole family or village
Narrative	• a project to study populations using mobile phone data is likely animated by a narrative that celebrates the ability and innovations of using readily available data in place of the painstaking process of data collection involved in conducting a census

I chose the case study involving collecting census data using mobile phone data. I chose this case study because it seemed to present some interesting challenges both logistically and ethically. Logistically, there would have to be a lot of assumptions made specifically concerning active mobile phone accounts per capita in a particular region. How would one even obtain such data? If you had a mechanism in place to determine such data, why could that mechanism not be used to collect the needed data? This would fall under the lens of Sociotechnical Systems along with other factors that make assumptions on how the systems producing the data are related to and “intertwined” with the people represented in the data.

Ethical Matrix: BMI + AI Tax	Government	Individuals Who Lost Job To AI
• How will each stakeholder benefit? • How might they each be harmed? • Are there less risky alternatives that could provide the same outcome?	• Increased tax on businesses • Increased burden on social service infra. • Strict AI-employment legislation	• Compensated for lost work • Lack of drive/opportunity • Retrain in AI-related field
• Will each stakeholder have a choice to use or be subject of the use? • Will they be able to determine how it is used with respect to their interests? • What will informed consent look like?	• The government will have a choice to not compensate workers • TBD • People will be begging for help	• They could choose to retrain • Not really • TBD